

## MAGNETIC FIELDS – CHAPTER 22

1. A magnetic field may exist at a point as a result of moving charged particle
2. When a tiny bar magnet is suspended horizontally from its center, it lines up along the north – south direction of earth's magnetic field.
3. The direction of the tiny magnet can be represented by so called magnetic field lines (magnetic flux).
4. In a uniform field, the lines are evenly spaced. The field is strong if the lines are close to each other.
5. Sketch field lines for (all sketches must have plane view and side view)
  - i) Straight conductor carrying current.
  - ii) A loop carrying current
  - iii) Solenoid with and without an iron core inside.

### Magnetic field

- A region of space where a magnetic material experiences a force.

Electromagnetic waves can travel in vacuum because electric field and magnetic field can recreate each other so no medium is needed.

The greater the distance from current-carrying conductor, the magnetic field strength decreases, so the separation of magnetic field lines increases.

In magnetic material, the atoms oscillate randomly, creating domain. When magnetic force is applied, the domain can align to concentrate the magnetic field.

In non magnetic material, the domains cancel off each other. When magnetic field is applied, the domain are not aligned.

Iron core is used in solenoid because it concentrates the magnetic field lines so that the magnetic force increases.

Soft iron core loses its magnetism, while hard iron core remains magnetised for a longer time.

## ELECTROMAGNETISM – CHAPTER 22

### A. FORCE ON A CURRENT-CARRYING CONDUCTOR

1. When two permanent magnets are placed close together with opposite poles facing each other, a uniform field (magnetic) can be produced.
2. A current-carrying conductor also produces a magnetic field around it when current is flowing through it.
3. When these two fields are put together at a point, a force is produced. This is due to interaction between the same types of field.
4. The direction of the force can be determined by using Fleming's Left Hand Rule.  
<http://www.youtube.com/watch?v=xdZsiBwkmf0>
5. Experiment shows that the size of the force acting on the current-carrying conductor is directly proportion to:
  - The **magnitude of current (I)**
  - The **magnitude of magnetic field strength (B)**
  - The **length of conductor placed in magnetic field (L)**

Formulae  $F = BIL$

6. To produce maximum force on the current-carrying conductor, the conductor should be placed at perpendicular to the magnetic field.
7. The equation can be re-write as

$$F = BIL \sin \theta$$

8. By re-arranging the equation

$$B = F / (IL)$$

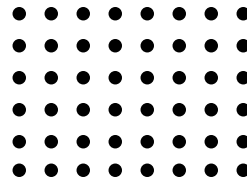
9. Magnetic flux density,  $B$  is defined as Force experienced per unit length by a long straight conductor carrying unit current and placed at right angles to the field at the point in space.
10. 1 Tesla is stated as The magnetic flux density where a wire carrying a current of 1A placed at right angles to the magnetic field experiences a force of 1N per metre of its length.
11. Experiment to measure the flux density of a magnetic field. **CURRENT BALANCE.(to be carried out in the lab)**

1. Write down the equation defining magnetic field density in terms of  $F$  the force it produce on a long, straight conductor of length  $l$  at an angle  $\theta$  to the field. Draw a clear diagram to illustrate the direction of the force relative to the current and magnetic field.
2. Define Tesla.
3. A wire carrying a current of 6.8 A is placed into a uniform magnetic field of strength of  $B = 0.40$  T at an angle of  $50^\circ$  to the field. Find the magnitude and direction of the resultant force on a 0.5 m length of the wire in the field.
4. A uniform metal rod of mass 50 g and length 0.20 m, carrying a current of 10 A is suspended horizontally at its two ends by two vertical springs. A horizontal magnetic field is to be applied perpendicular to the rod so that the spring is not extended.
  - a) draw a sketch diagram showing the direction of the magnetic field relative to the direction of the current.
  - b) calculate the magnitude of the magnetic flux density.
5. A current carrying superconducting wire, made of niobium of density  $8600 \text{ kgm}^{-3}$  and of radius 0.100 cm, is placed at right angle in a horizontal magnetic field of flux density 2.0 mT. The wire just levitates, that is, floating in the air.
  - a) Sketch a diagram to show the relative direction of the magnetic field and the current.
  - b) what is the current flowing in the wire?

## B. FORCE ON A MOVING CHARGE

1. A current-carrying conductor experiences a force when it is placed in a magnetic field.
2. Since current is flow of electrons, the force acting on the conductor is the resultant forces acting on each of the moving electrons (or any other charge particles).
3. Diagram and derivation of the formulae.

X X X X X X X X  
 X X X X X X X X  
 X X X X X X X X  
 X X X X X X X X  
 X X X X X X X X



[http://www.youtube.com/watch?v=-ZL-hAfly\\_Y&NR=1](http://www.youtube.com/watch?v=-ZL-hAfly_Y&NR=1)

4. The direction of the force can be determined by Fleming's Left Hand Rule
5. If the charged particle enters the magnetic field at an angle of  $\theta$ , the force, \_\_\_\_\_

$$F = Bqv \sin \theta$$

6. The relation between  $F_B$  and  $F_C$

$$\pi = p i$$

[http://www.youtube.com/watch?v=a2\\_wUDBl-g8&NR=1](http://www.youtube.com/watch?v=a2_wUDBl-g8&NR=1)

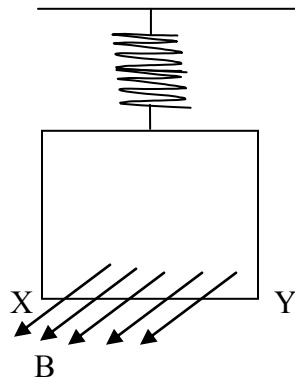
Magnetic force  
 $= BIL$   
 $= B (q/t) L$   
 $= Bq (L/t)$   
 $= Bqv$

Magnetic force = Centripetal force  
 $Bqv = m (v^2) / r$   
 $Bq = mv / r$   
 $r = (mv) / (Bq)$

$v = 2 (\pi) r / T$   
 $r = m (2 \pi r / T) / (Bq)$   
 $BqT = 2 (\pi) m$   
 $T = (2 \pi m) / (Bq)$

Centripetal force  
 $= m (v^2) / r$

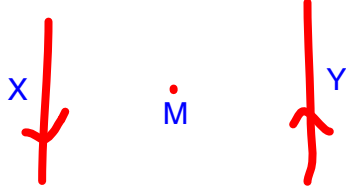
1. An electron with velocity of  $3.5 \times 10^7 \text{ ms}^{-1}$  enters at a right angle into a magnetic field of flux density of 0.50 mT. What will be the radius of the path of the electron while in the field?
2. An electron is accelerated from rest by a potential difference of 12 kV. It then enters at right angle into a uniform magnetic field of strength 100 mT. Calculate
  - a) the speed of the electron on the entering the field.
  - b) The radius of its circular path in the magnetic field.
3. A helical spring supports a rectangular coil carrying a current as shown in the figure. The lower part XY of the coil is right angle to a uniform magnetic field but there is no field elsewhere. The coil has 100 turns, carries a current 1.0 mA and has a mass of 10 g ; the length of XY is 5.0 cm. When the current in the coil is switched off, the coil oscillates with a period of 1.0 s. Finally, stops 1.0 mm above its initial position. Calculate the magnetic flux density of the field.



### D. Forces Between current-carrying conductor

1. Consider two wires X and Y carrying current  $I_x$  and  $I_y$ .

Diagram of 2 wires



2. By using Right-hand grip rule, the magnetic field due to  $I_x$  at M, is directed towards out of page.
3. Similarly at M, due to  $I_y$  is also towards out of page.
4. The two forces at point M, is the sum of forces due to  $I_x$  and  $I_y$ .
5. The force perunit length ( $F/l$ ) acting at point M or at any other points is given by ;

$$\frac{F}{l} = \frac{\mu_o I_x I_y}{2\pi r}$$

$\mu_o$  – permeability of free space  
 $4\pi \times 10^{-7} \text{ Hm}^{-1}$

7. At point N, equal distance from the wires X and Y, if  $I_x$  and  $I_y$  is same magnitude, the resultant force at this point is zero.

1. a) i) Define magnetic flux density.

.....  
 .....  
 ..... [2]

- ii) State the SI unit for magnetic flux density.

..... [1]

- b) Sketch the magnetic flux pattern produced by a solenoid when a current flows in it.

- c) A charged particle enters perpendicularly into a magnetic field. State and explain the path traced by the charged particle.

.....  
 .....  
 .....  
 ..... [2]

**2a)** Define *magnetic flux density* in terms of force on a conductor carrying a current.

.....  
.....  
.....[2]

**(b)** Figure 4.1 below shows a long straight wire X carrying a current  $I_X$  flowing upwards.

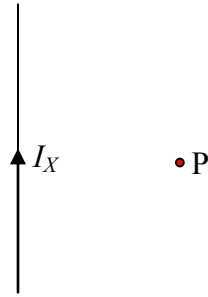


Fig 4.1

In the space below sketch the magnetic field lines as seen from the top. Indicate the direction of the field lines. [1]

A second long straight wire Y carrying a current in the opposite direction is then placed at the point P parallel to the wire X. In the space below, indicate the direction of the magnetic field on wire Y at P as seen from the top. Indicate also the direction of the force acting on it. [2]

**(c)** Hence, explain how you deduce the direction of the force on the wire X.

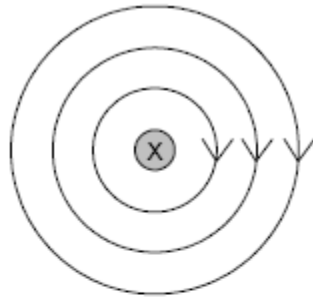
.....  
.....[1]

1. A straight conductor carries current into the page as shown.

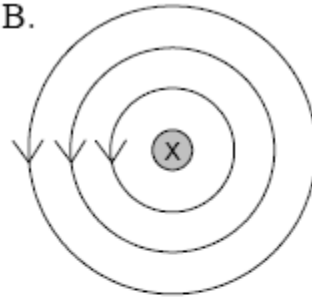


Which **one** of the following diagrams best represents the magnetic field pattern around the conductor?

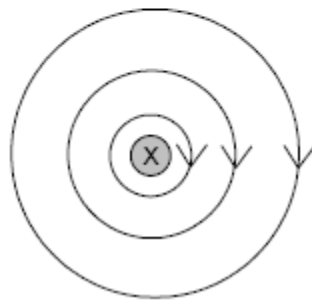
A.



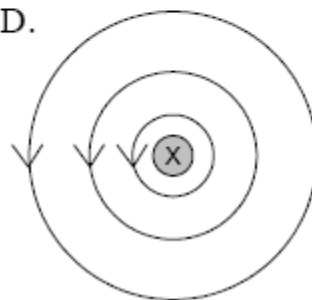
B.



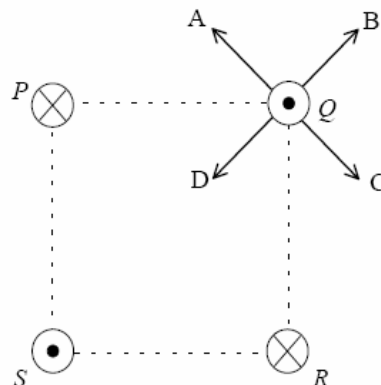
C.



D.



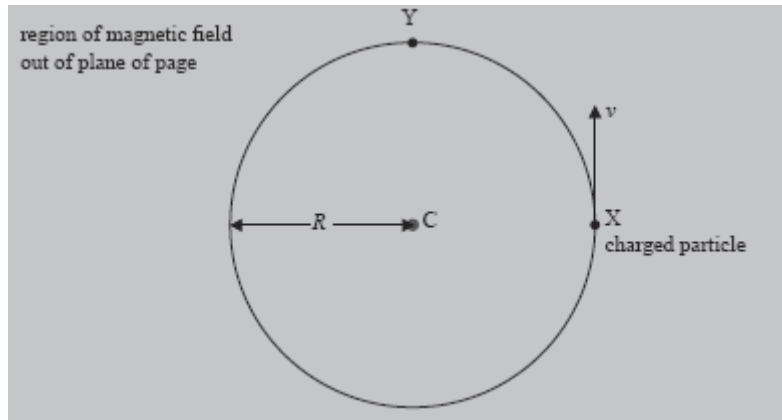
2. In the diagram four long wires are placed at the corner of a square and carry equal currents. The direction of the current in wires P and R is into the plane of the paper and in wires Q and S is out of the plane of the paper.



Which labelled arrow correctly shows the direction of the resultant force on wire Q?

- A. Arrow A      B. Arrow B      C. Arrow C      D. Arrow D

3. A charged particle is projected from point X with speed  $v$  at right angles to a uniform magnetic field. The magnetic field is directed out of the plane of the page. The particle moves along a circle of radius  $R$  and centre C as shown in the diagram below.

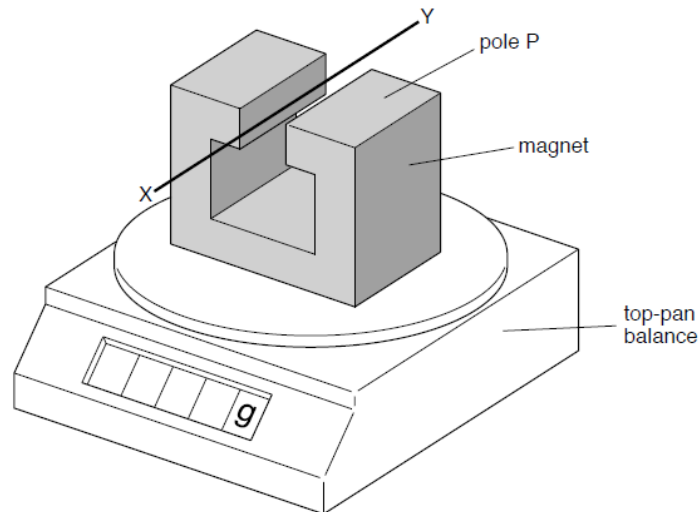


- (a) On the diagram above, draw arrows to represent the magnetic force on the particle at position X and at position Y. [1]
- (b) State and explain whether the charge is positive **or** negative. [1]
- (c) A second identical charged particle is projected at position X with a speed  $\frac{v}{2}$  in a direction opposite to that of the first particle. On the diagram above, draw the path followed by this particle. [2]
- (d) If the charge particle is an electron entering into a magnetic field with 0.02 T with a speed of  $2 \times 10^5 \text{ ms}^{-1}$ . Determine
- the radius of the circle.
  - time taken to complete the circle.
  - show that the period is independent of radius. [9]

4. (a) Define the *tesla*.

[3]

- (b) A large horseshoe magnet produces a uniform magnetic field of flux density  $B$  between its poles. Outside the region of the poles, the flux density is zero. The magnet is placed on a top-pan balance and a stiff wire XY is situated between its poles, as shown in Fig. 3.1.



**Fig. 3.1**

The wire XY is horizontal and normal to the magnetic field. The length of wire between the poles is 4.4 cm.

A direct current of magnitude 2.6 A is passed through the wire in the direction from X to Y.

The reading on the top-pan balance increases by 2.3 g.

- (i) State and explain the polarity of the pole P of the magnet.

[3]

- (ii) Calculate the flux density between the poles.

[2]

- (c) Explain how the readings top-pan balance will change when a low frequency sinusoidal current with maximum current as above.

.....[2]

**[Total marks 10 M]**

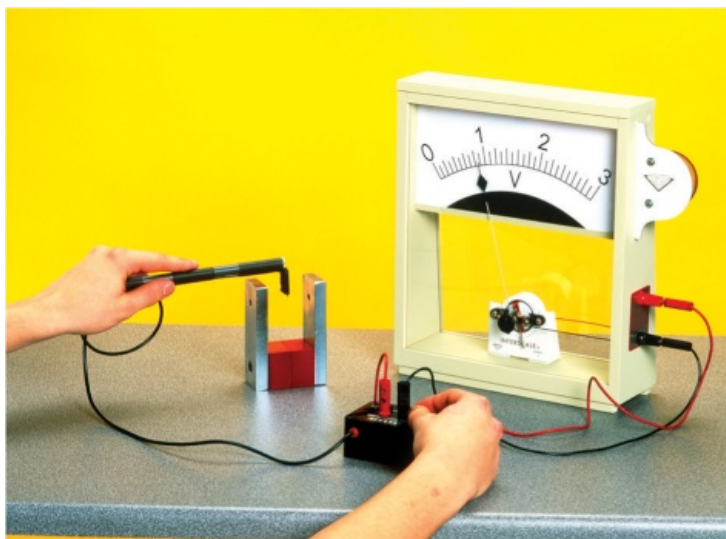
## 24.4 Measuring magnetic flux density

Practical Activity 24.2 looks at two practical methods for measuring magnetic flux density.

### PRACTICAL ACTIVITY 24.2 MEASURING MAGNETIC FLUX DENSITY

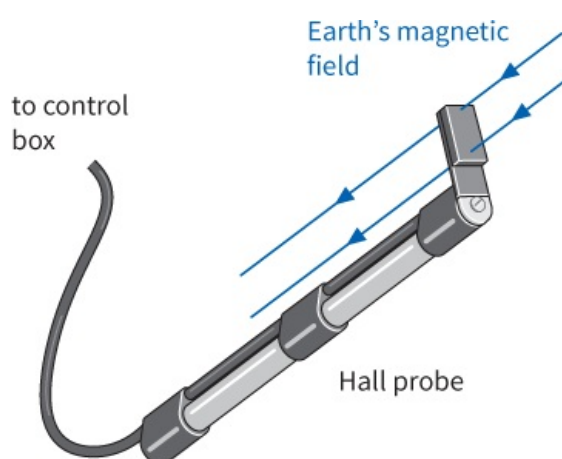
#### Measuring $B$ with a Hall probe

The simplest device for measuring magnetic flux density  $B$  is a Hall probe (Figure 24.12). When the probe is held so that the field lines are passing at right angles through the flat face of the probe, the meter gives a reading of the value of  $B$ . Some instruments are calibrated so that they give readings in microteslas ( $\mu\text{T}$ ) or milliteslas (mT). Others are not calibrated, so you must either calibrate them or use them to obtain relative measurements of  $B$ .



**Figure 24.12:** Using a Hall probe to measure the flux density between two magnets.

A Hall probe must be held so that the field lines are passing directly through it, at right angles to the flat surface of the probe (Figure 24.13). If the probe is not held in the correct orientation, the reading on the meter will be reduced.

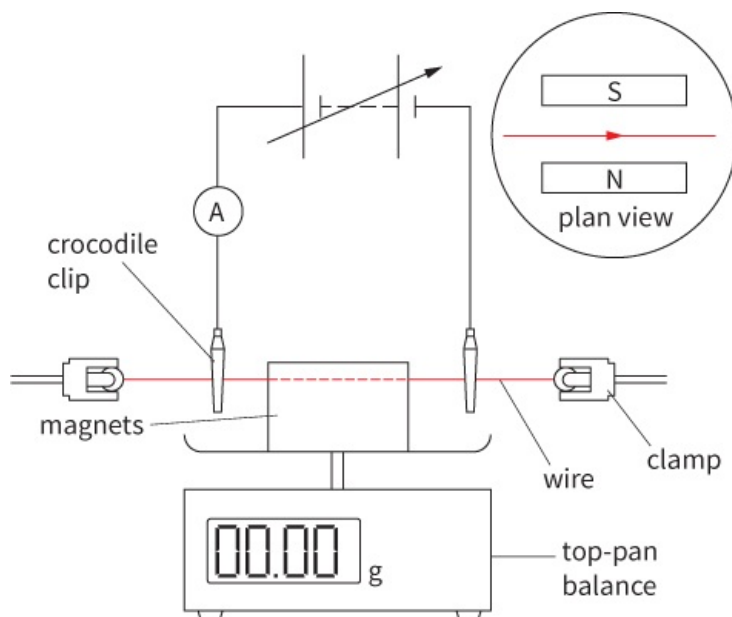


**Figure 24.13:** Magnetic flux lines must pass through the probe at  $90^\circ$  to the surface.

A Hall probe is sensitive enough to measure the Earth's magnetic flux density. The probe is first held so that the Earth's field lines are passing directly through it, as shown in Figure 24.13. In this orientation, the reading on the voltmeter will be a maximum. The probe is then rotated through  $180^\circ$  so that the magnetic field lines are passing through it in the opposite direction. The change in the reading of the meter is twice the Earth's magnetic flux density. There is more about how the Hall probe works in [Chapter 25](#).

## Measuring $B$ with a current balance

Figure 24.14 shows a simple arrangement that can be used to determine the flux density between two magnets. The magnetic field between these magnets is (roughly) uniform. The length  $L$  of the current-carrying wire in the uniform magnetic field can be measured using a ruler.



**Figure 24.14:** An arrangement to determine magnetic flux density in the laboratory.

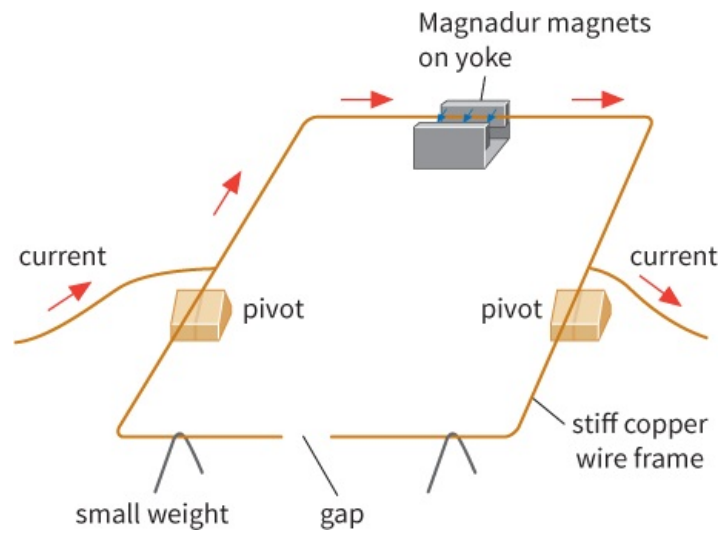
When there is no current in the wire, the magnet arrangement is placed on the top pan and the balance is zeroed. Now, when a current  $I$  flows in the wire, its value is shown by the ammeter. The wire experiences an upward force and, according to Newton's third law of motion, there is an equal and opposite force on the magnets. The magnets are pushed downwards and a reading appears on the scale of the balance. The force  $F$  is given by  $mg$ , where  $m$  is the mass indicated on the balance in kilograms and  $g$  is the acceleration of free fall ( $9.81 \text{ m s}^{-2}$ ).

Knowing  $F$ ,  $I$  and  $L$ , the magnetic flux density  $B$  between the magnets can be determined using the equation:

$$B = \frac{F}{IL}$$

You can also use the arrangement in Figure 24.14 to show that the force is directly proportional to the current.

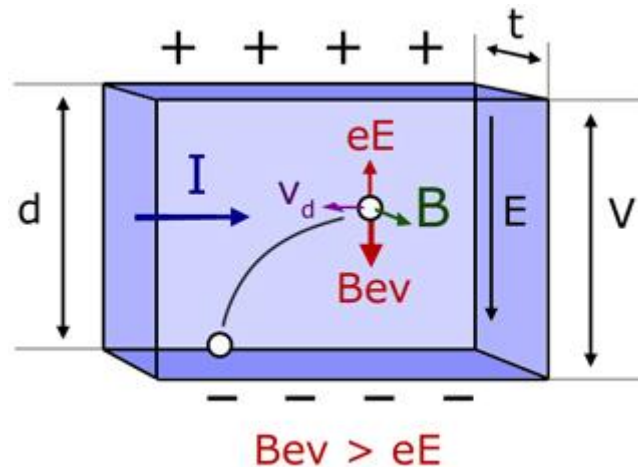
A system like this in effect 'weighs' the force on the current-carrying conductor, and is an example of a current balance. Another version of a current balance is shown in Figure 24.15. This consists of a wire frame that is balanced on two pivots. When a current flows through the frame, the magnetic field pushes the frame downwards. By adding small weights to the other side of the frame, you can restore it to a balanced position.



**Figure 24.15:** A simple laboratory current balance.

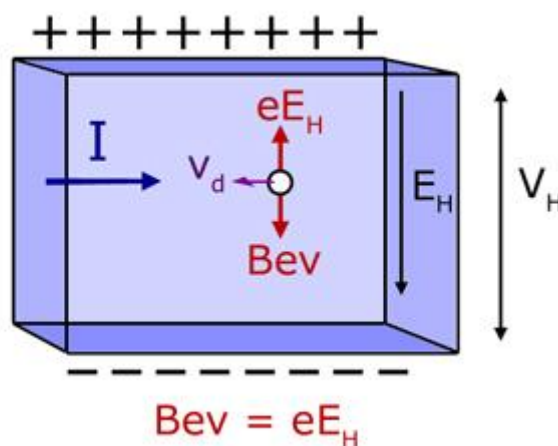
## The Hall Effect

The Hall Effect refers to the potential difference that builds up across opposing faces of a cuboidal shaped material when it is carrying current and placed in a uniform magnetic field.



The potential difference  $V$  is a consequence of free electrons in the material being deflected by the magnetic field. The electrons move in arcs and build up on the bottom face. Removal of free electrons from the top half of the solid makes the top face positive.

The electric field  $E$  produced by the deposited electrons is very weak to start with. As more electrons accumulate, it grows stronger. The resultant force on electrons ( $Bev - eE$ ) becomes weaker and weaker. Eventually a point is reached when the magnetic and electric forces are equal ( $Bev = eE$ ) and no more electrons are added.



When this happens the potential difference across the top and bottom faces is a maximum ( $V_H$ ). The p.d. is called the Hall Voltage and is a measure of the magnetic field passing through the solid.

An expression for the Hall Voltage can be obtained from a consideration of the forces on an electron, when they are balanced.

The downward magnetic force  $F$  is given by :

$$F = Bev$$

The upward electric force is given by :

$$F = eE$$

For a uniform electric field,

$$E = \frac{V_H}{d}$$

where  $d$  is the thickness of the sample.

Substituting for  $E$  into the electric force equation,

$$F = \frac{eV_H}{d}$$

Equating the magnetic force with the electric force :

$$Bev = \frac{eV_H}{d}$$

cancelling the  $e$ 's and making  $V_H$  the subject of the equation,

$$V_H = Bvd$$

Quoting the **Drift Velocity** equation

$$I = nAve$$

Rearranging to make the drift velocity  $v$  the subject,

$$v = \frac{I}{nAe}$$

Now, substituting for  $\mathbf{v}$  into the equation for  $\mathbf{V_H}$  ,

$$V_H = \frac{BdI}{nAe}$$

The cross-sectional area  $\mathbf{A}$  of the sample between left and right faces is simply the product of the height  $\mathbf{d}$  and its thickness  $\mathbf{t}$ .

$$A = dt$$

Substituting for  $\mathbf{A}$  in the new expression for  $\mathbf{V_H}$  ,

$$V_H = \frac{BdI}{ndte} = \frac{BI}{nte}$$

Shuffling around bottom terms to make the equation more memorable :

$$\underline{V_H = \frac{BI}{net}}$$

The Hall voltage is larger in semiconductors than metals. This is because  $\mathbf{V_H}$  is inversely proportional to the number of charge carriers/unit volume  $\mathbf{n}$ . Values for  $\mathbf{n}$  are much smaller in semiconductors. So  $\mathbf{V_H}$  is larger.

Typical values of  $\mathbf{V_H}$  for specimens under similar conditions :

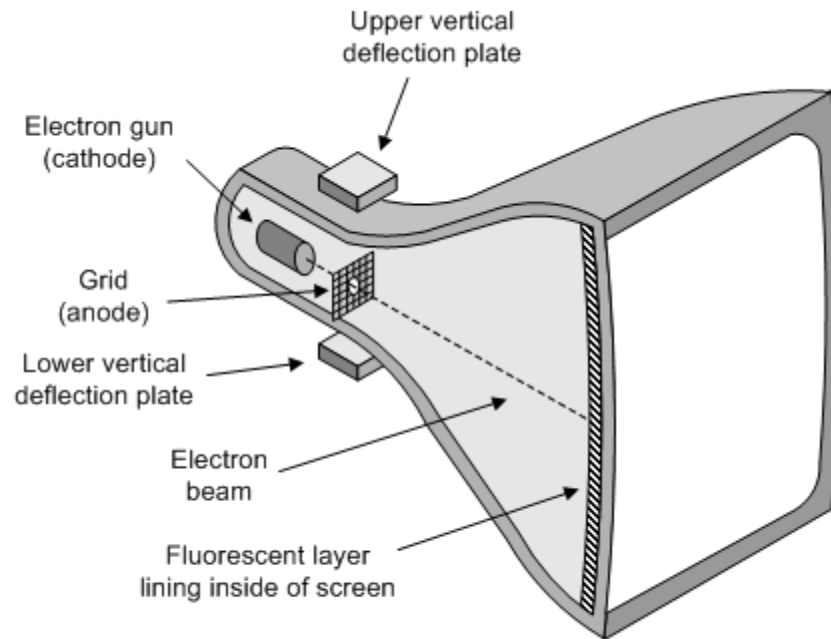
metals  $\sim 10^{-6}$  V

semiconductors  $\sim 10^{-2}$  V

A useful application of the Hall Effect is to test whether a material is p or n-type semiconductor.

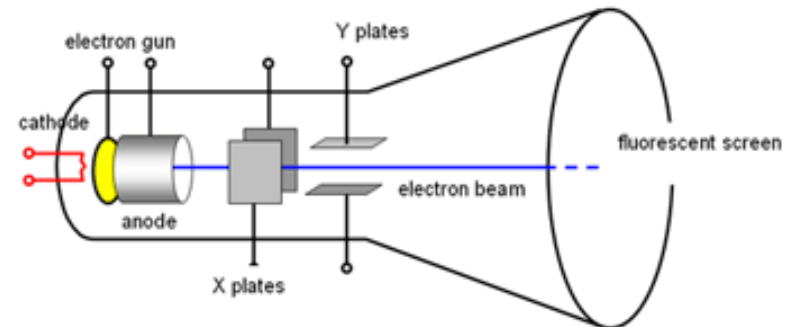
Note also how Hall Voltage  $\mathbf{V_H}$  is inversely proportional to sample thickness  $\mathbf{t}$ . So a thin sample (a small  $\mathbf{t}$ ) will give a large Hall Voltage.

# Production of Electron Beam

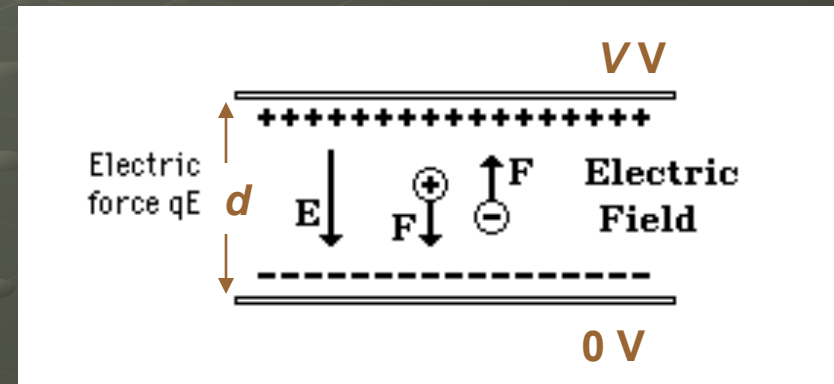
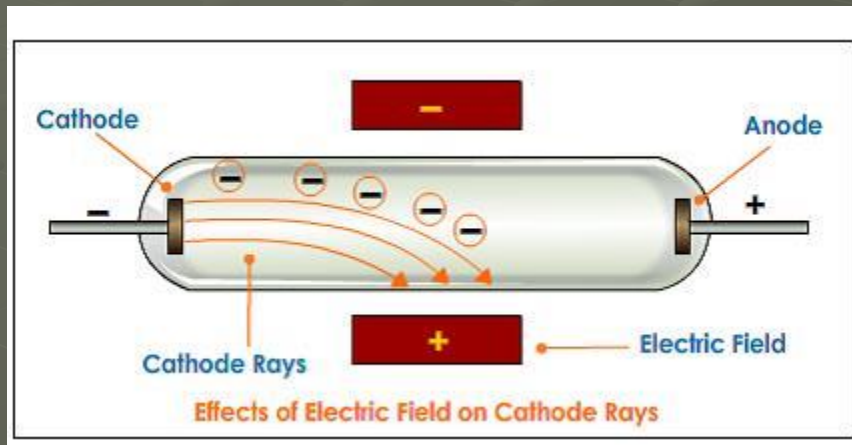


**Cross-sectional representation of a cathode ray tube**

## Cathode Ray Tube



# Electrons in an Electric Field

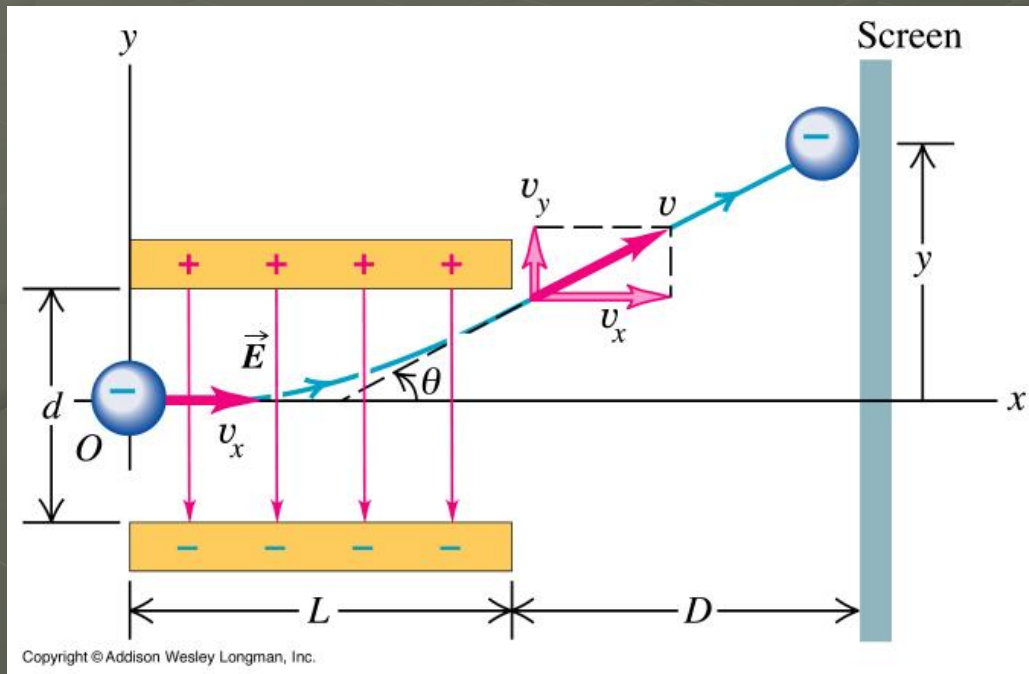


Electric Force,  $F_e = e E$

$$ma = e (V / d)$$

$$a = e V / m d$$

# Deflection of Electrons in an Electric Field



Apply dynamic equations to motion of charged particles in an electric field.

Its verticle displacement:

$$y = \frac{1}{2} at^2 = \frac{1}{2} \left( \frac{eV}{md} \right) \frac{x^2}{v^2} = \left( \frac{eV}{2mdv^2} \right) x^2$$

Its verticle speed:

$$v_y = at = \left( \frac{eV}{md} \right) \left( \frac{x}{v} \right) = \frac{eVx}{mdv}$$

At any point:

The direction of the beam:

$$\tan \theta = v_y / v$$

The speed of the beam:

$$v_r = \sqrt{v^2 + v_y^2}$$

# Example

A beam of electrons travelling at  $1.35 \times 10^7 \text{ m s}^{-1}$  enters a uniform electric field between two plates of length  $l$ , separated by a distance  $d$ . One plate is at a potential of  $+50 \text{ V}$  and the other is at  $-50 \text{ V}$ , as shown in Fig 27.8. If  $l = 0.060 \text{ m}$  and  $d = 0.020 \text{ m}$ , find  $\theta$ , the angular deflection of the beam.

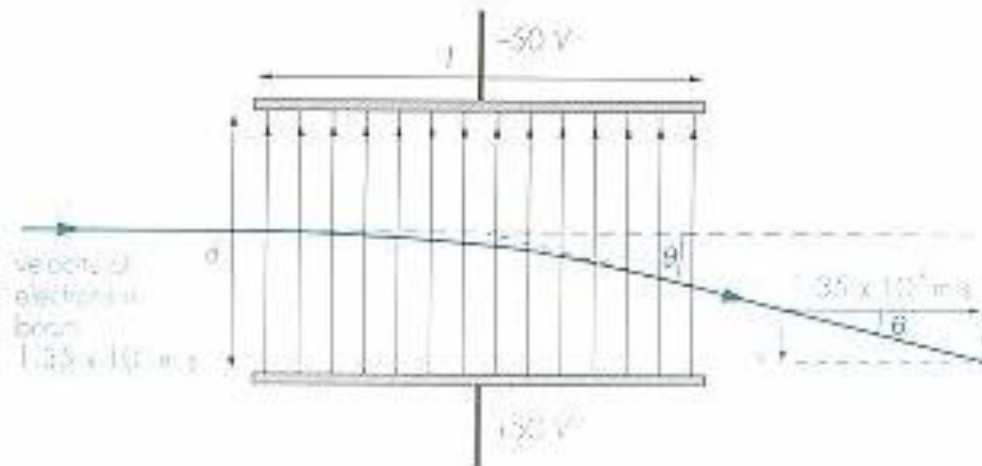


FIG 27.8 Finding the deflection of the electron beam in Example 27.2

# Solution

The electric field,  $E$ , between the plates is the potential gradient

$$E = \frac{100 \text{ V}}{0.02 \text{ m}} = 5000 \text{ V m}^{-1}$$

The field is in a direction towards the top of the page, so an electron, which has a negative charge, will have a force on it acting downwards towards the bottom of the page. Since this force is always downwards, there is no horizontal acceleration, and the horizontal velocity is therefore constant. This gives the time taken to travel between the plates as

$$\text{time taken} = \frac{\text{distance}}{\text{velocity}} = \frac{0.060 \text{ m}}{1.35 \times 10^7 \text{ m s}^{-1}} = 4.44 \times 10^{-9} \text{ s}$$

The downward force on an electron is given by

$$\begin{aligned}\text{force downwards} &= Ee \\ &= 5000 \text{ V m}^{-1} \times 1.60 \times 10^{-19} \text{ C} \\ &= 8.0 \times 10^{-16} \text{ N}\end{aligned}$$

While between the plates, the downward acceleration is therefore given by

$$\begin{aligned}\text{acceleration} &= \frac{\text{force downwards}}{\text{mass}} = \frac{8.0 \times 10^{-16} \text{ N}}{9.11 \times 10^{-31} \text{ kg}} \\ \text{downwards} &= 8.78 \times 10^{14} \text{ m s}^{-2}\end{aligned}$$

(Note how huge this acceleration is. This is possible because of the very small mass of the electron.)

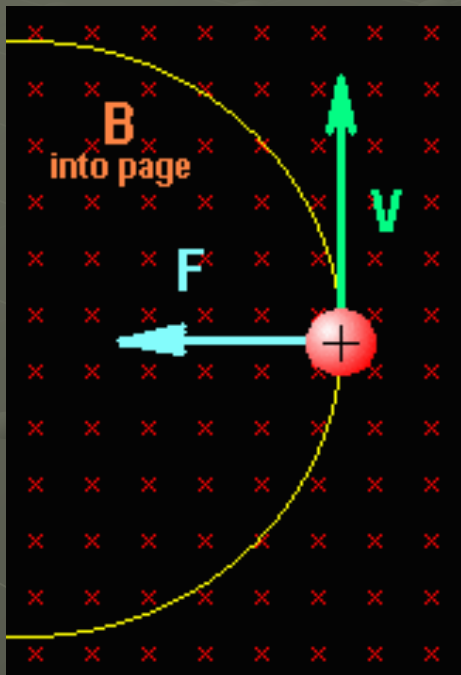
For the downward direction, it is now possible to use  $v = u + at$  to get

$$\begin{aligned}v &= 0 + (8.78 \times 10^{14} \text{ m s}^{-2} \times 4.44 \times 10^{-9} \text{ s}) \\ &= 3.90 \times 10^6 \text{ m s}^{-1}\end{aligned}$$

The forward velocity is unchanged, so using the vector diagram shown in the bottom right part of Fig 27.8 gives

$$\begin{aligned}\tan \theta &= \frac{v}{1.35 \times 10^7 \text{ m s}^{-1}} = \frac{3.90 \times 10^6 \text{ m s}^{-1}}{1.35 \times 10^7 \text{ m s}^{-1}} \\ &= 0.289 \\ \theta &= 16^\circ \text{ to 2 sig. figs}\end{aligned}$$

# Electrons in a Magnetic Field



Magnetic Force,  $F = Bev$

As it is making a circular motion:

Centripetal force = Magnetic force

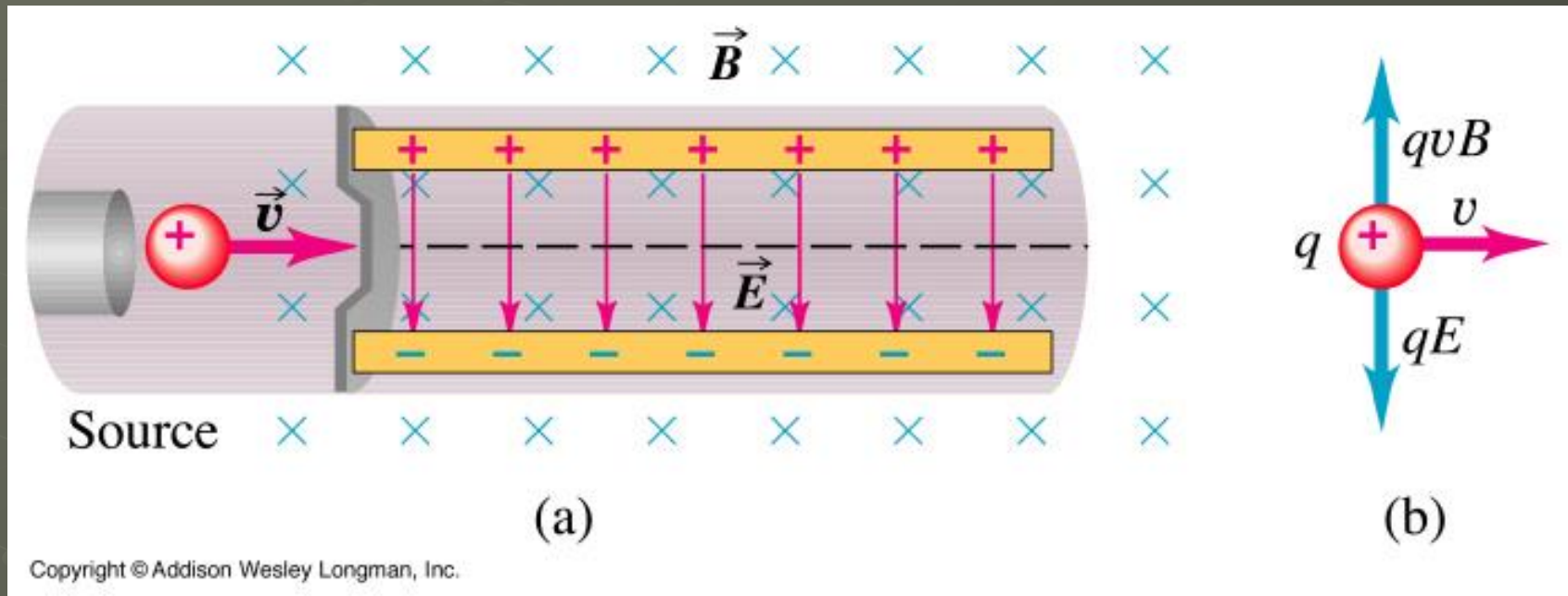
$$mv^2 / r = Bev$$

$$v = Ber / m$$

OR

$$r = mv / Be$$

# Combined Electric & Magnetic Fields



Particles with the right speed will move undeflected in the combined electric and magnetic fields.

For zero deflection:

Magnetic force = Electric force

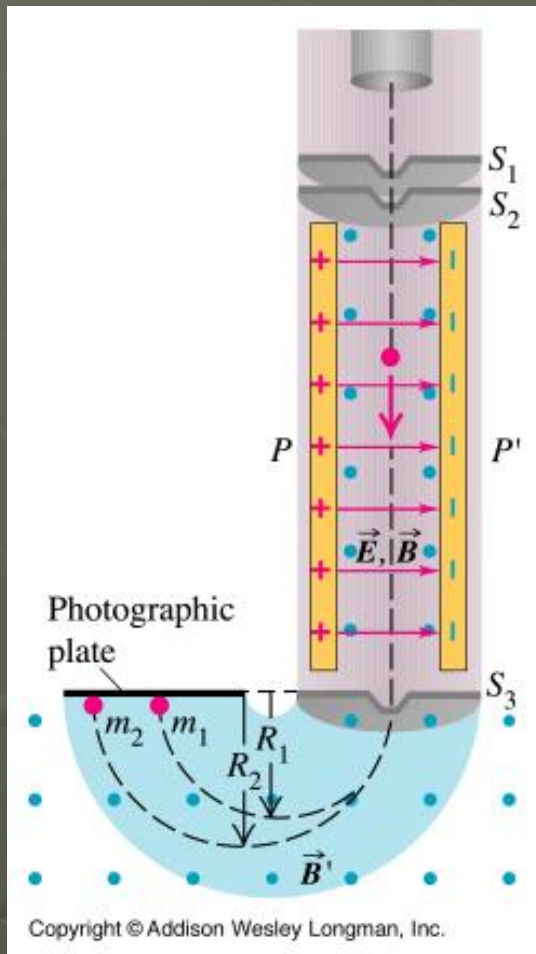
$$Bev = eE$$

$$v = E / B$$

OR

$$v = V / Bd$$

# Specific Charge, $e/m$



Mass Spectrometer

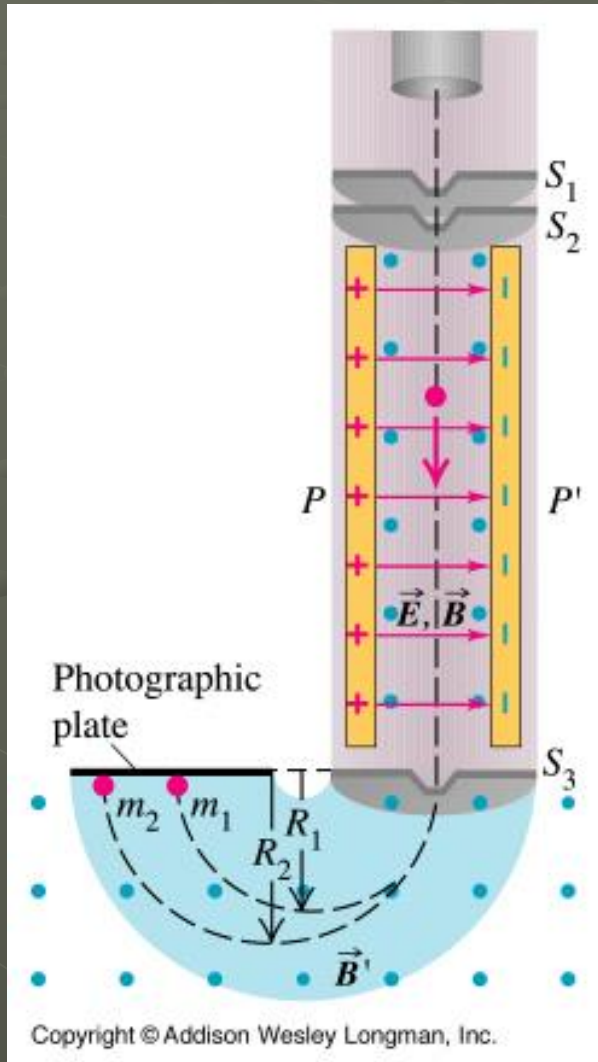
If a particle carries a charge of  $e$  and has a mass  $m$ .

Then the ratio  $e/m$  is the specific charge of the particle.

It is measured in  $C\ kg^{-1}$

To find this value we use a mass spectrometer as shown.

# Determining $e/m$



**The Electron Gun:** Produce the electrons and accelerate them from the cathode to the anode

$$\frac{1}{2} mv^2 = eV$$

$$v = \sqrt{\frac{2eV}{m}}$$

**The Velocity Selector:** Electrons with the right speed will move undeflected in the combined electric and magnetic fields and will pass through a gap  $S_3$ .

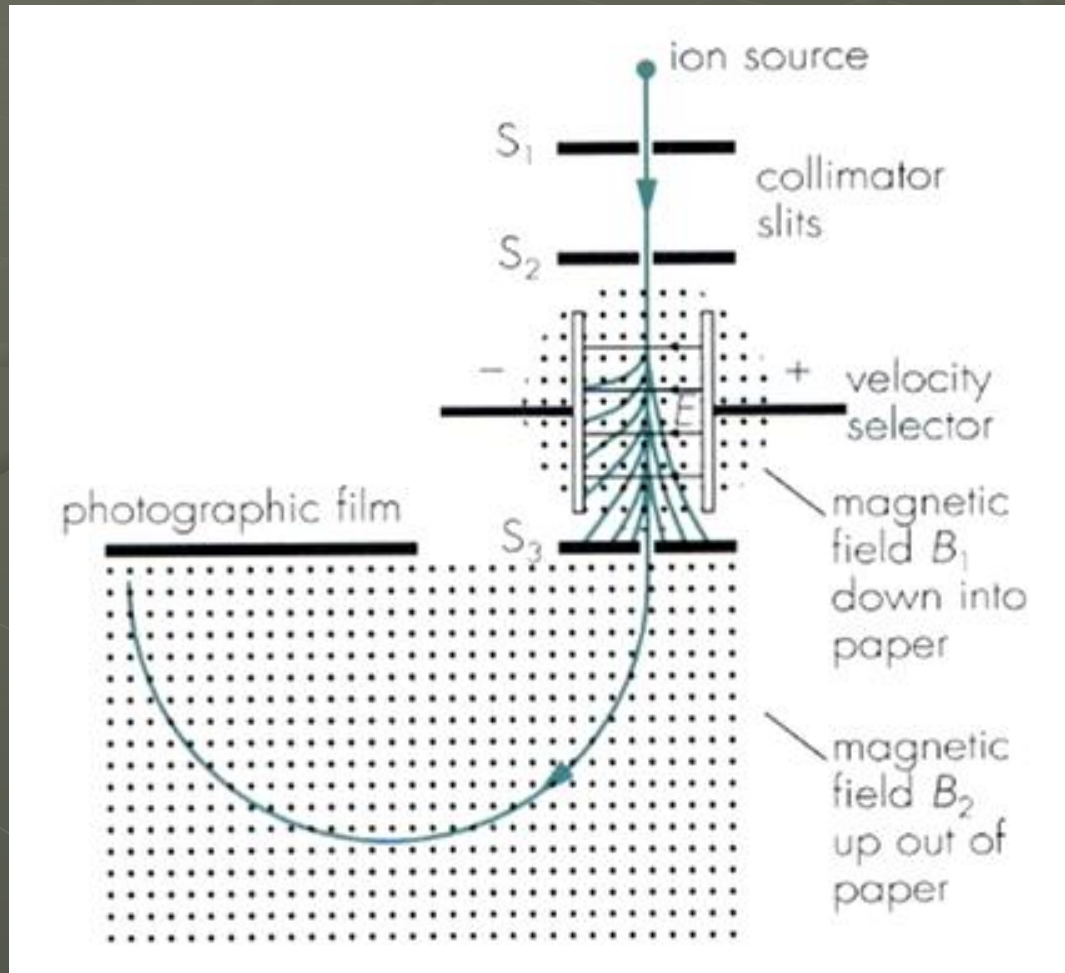
$$v = \frac{E}{B}$$

**The Deflector:** Electrons emerging from slit  $S_3$  is deflected through a magnetic field. They are made to strike a photographic plate and the radius of the circular motion measured.

$$\frac{mv^2}{r} = B_2 ev$$

$$\frac{e}{m} = \frac{v}{B_2 r}$$

# Determining $e/m$ (1)



## 1. Ionisation:

### Thermionic emission:

Current is supplied to a cathode (tungsten wire) and heating it up to produce electron beam.

## 2. Acceleration

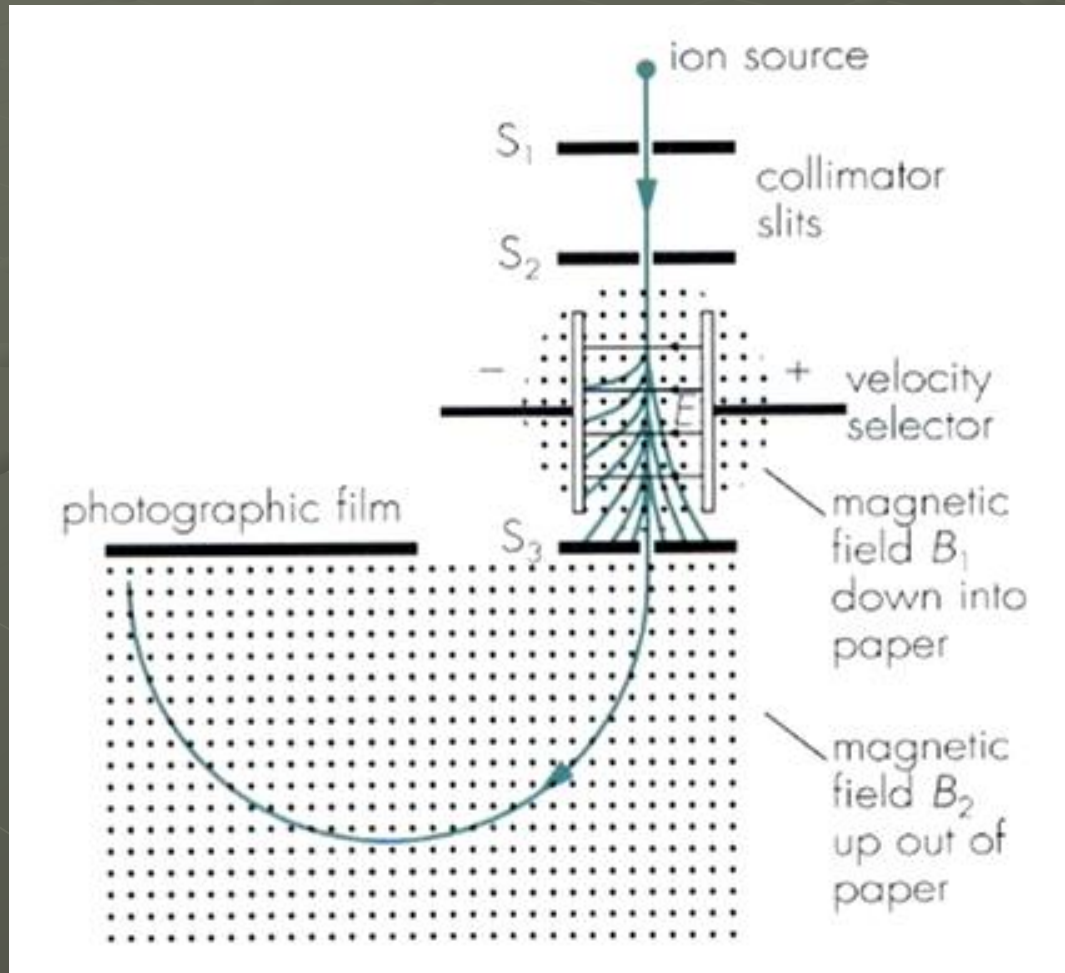
The plates are connected to an accelerating +ve voltage  $V$  to accelerate the electrons through slit  $S_1$  and  $S_2$  into a velocity selector of cross  $E$  &  $B$  fields.

$$\frac{1}{2} mv^2 = eV \quad \text{--- (1)}$$

$$v = \sqrt{\frac{2eV}{m}} \quad \text{--- (2)}$$

All electrons have the same K.E.

# Determining (2)



### 3. Deflection:

Only those with speed

$$v = E / B$$

will move undeflected in the cross  $E$  &  $B$  fields, passing through slit  $S_3$ .

### 4. Detection:

The selected electrons are acted on by  $F_{em}$  and follow a circular path of radius  $r$  in the magnetic field  $B_2$ .

The radius can be measured as the electrons darken the photographic plate where they strike.

$$B e v = m v^2 / r \quad \text{--- (3)}$$

$$e / m = v / (B r) \quad \text{--- (4)}$$

# The electron-volt (eV)

Find the kinetic energy that an electron acquires if it falls through a potential difference of 1 V, and its charge is  $-1.60 \times 10^{-19} \text{ C}$ .

This can be found by recalling that

one volt = one joule per coulomb =  $1 \text{ J C}^{-1}$

Therefore for a charge of  $-1.60 \times 10^{-19} \text{ C}$ , the kinetic energy  $E_k$  acquired as the potential difference rises by one volt is

$$E_k = 1 \text{ J C}^{-1} \times 1.60 \times 10^{-19} \text{ C} = 1.60 \times 10^{-19} \text{ J}$$

$$1.60 \times 10^{-19} \text{ J} \\ = 1 \text{ electron-volt} \\ (\text{eV})$$

# Exercise

An electron is accelerated by a p.d. of 500 V. Calculate the gain in kinetic energy

(i) in joules

$$8.0 \times 10^{-17} \text{ J}$$

(ii) in electron-volt

$$500 \text{ eV}$$

# Exercise

An electron leaves a cathode at zero potential and travels through a vacuum to an anode at + 200 V. Calculate

(i) the kinetic energy that it acquires.

$$3.2 \times 10^{-17} \text{ J}$$

(ii) the speed when it reaches the anode.

$$8.4 \times 10^6 \text{ m s}^{-1}$$

# Example

- A hydrogen has a charge-to-mass ratio of  $q/m = 9.65 \times 10^7 \text{ C kg}^{-1}$ . A Bainbridge mass spectrometer has  $B_1 = 0.93 \text{ T}$ ,  $B_2 = 0.61 \text{ T}$  and  $E = 3.7 \times 10^6 \text{ V m}^{-1}$ . Calculate the radius of the paths of each of the following ions in the mass spectrometer:

(a)  $\text{H}^+$

0.0676 m

(b)  $\text{He}^+$

0.270 m

(c)  $\text{He}^{2+}$

0.135 m

# Example 1

An electron gun operating at 3000 V is used to shoot electrons into the space between two oppositely charge parallel plates. The plate spacing is 50 mm and its length is 100 mm.

Calculate the deflection of the electrons at the point where they emerge from the field when the plate p.d. is 100 V. Assume the specific charge  $e/m$  for the electron is  $1.76 \times 10^{12} \text{ Ckg}^{-1}$

# Solution 1

*Solution*

(a) Use the velocity selector equation for the speed  $v$ .

$$v = \frac{V_p}{Bd} = \frac{720}{0.65 \times 10^{-3} \times 50 \times 10^{-3}} = 2.22 \times 10^7 \text{ m s}^{-1}$$

(b) Use the equation for vertical displacement  $y$  in an electric field.

$$y = \frac{eV_p x^2}{2mdv^2}$$

$$\text{Hence } \frac{e}{m} = \frac{2yd v^2}{V_p x^2}$$

$$\text{So } e/m = \frac{2 \times 25 \times 10^{-3} \times 50 \times 10^{-3} \times (2.22 \times 10^7)^2}{720 \times (100 \times 10^{-3})^2}$$

$$e/m = 1.71 \times 10^{11} \text{ C kg}^{-1}$$

# Example 2

A beam of electrons is fired into a uniform electric field as in Fig. 27.2F. The length of the plates is 100 mm and the plate spacing is 50 mm. When a constant p.d. of 720 V is applied across the deflecting plates, the beam is deflected vertically at the edge of the field by a distance of 25 mm. The beam is straightened out by applying a uniform magnetic field of strength of 0.65 mT perpendicular to the beam with the same plate p.d. Calculate

- (a) the speed of the beam entering the fields,
- (b) the specific charge of the electron.

# Solution 2

*Solution*

First, we need to calculate the speed  $v$  of the electrons from the electron gun.

$$\frac{1}{2}mv^2 = eV_A$$

$$\begin{aligned}\text{Hence } v^2 &= 2 \frac{e}{m} V_A = 2 \times 1.76 \times 10^{11} \times 3000 \\ &= 10.56 \times 10^{14} \text{ m}^2 \text{ s}^{-2} \\ v &= 3.25 \times 10^7 \text{ m s}^{-1}\end{aligned}$$

Now the deflection can be calculated using

$$y = \frac{(eV_p)}{2mdv^2} x^2$$

with  $x = 100 \text{ mm}$  and  $d = 50 \text{ mm}$

Thus

$$y = \frac{1.76 \times 10^{11} \times 1000 \times (100 \times 10^{-3})^2}{2 \times 50 \times 10^{-3} \times 10.56 \times 10^{14}} = 1.67 \times 10^{-2} \text{ m}$$

# Example 3

An electron gun operating at 2000 V is used to shoot electrons into a magnetic field of strength 1.9 mT. The initial direction of the beam is at right angles to the field direction.

- (a) Calculate the specific charge on an electron if the beam forms a circle of diameter 160 mm.
- (b) Describe the beam path if the initial direction of the beam is changed to  $30^\circ$  to the field direction.

# Solution 3

*Solution*

(a)  $V_A = 2000 \text{ V}$ ,  $B = 1.9 \times 10^{-3} \text{ T}$ ,  $r = 0.160/2 \text{ m}$

Hence

$$\frac{e}{m} = \frac{2VA}{B^2 r^2} = \frac{2 \times 2000}{(0.0019)^2 \times (0.080)^2} = 1.73 \times 10^{11} \text{ C kg}^{-1}$$

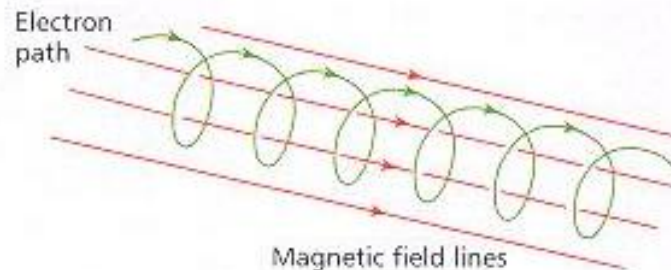


Fig. 27.2E Electron spirals

- (b) The electrons spiral round the field lines as shown in Fig. 27.2E. The component of velocity parallel to the field,  $v \cos 30^\circ$ , does not change because the magnetic force is always at right angles to the field direction. The perpendicular component,  $v \sin 30^\circ$ , is the speed at which the electrons 'circle' in the field. This is exactly half the value of the speed when the direction was at right angles. So the 'circle' radius is half the previous value, giving 40 mm. In other words,

(i) at right angles,  $v = Be r_1 / m$  where  $r_1 = 80 \text{ mm}$ ,

(ii) at  $30^\circ$ ,  $v \sin 30^\circ = Be r_2 / m$  where  $r_2$  is the spiral radius.

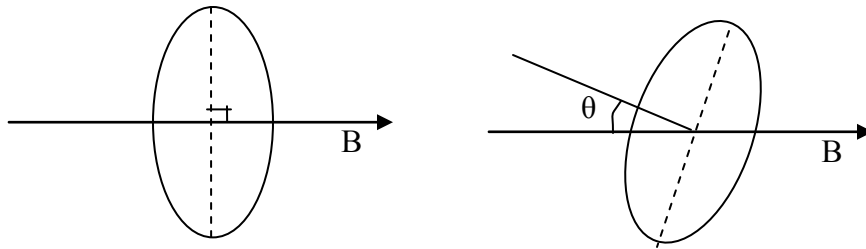
Hence  $r_2 = r_1 \sin 30^\circ = 40 \text{ mm}$ .

## ELECTROMAGNETIC INDUCTION – CHAPTER 23

### 1. Magnetic flux definition, $\Phi$ :

Product of magnetic flux density and perpendicular surface area.

SI unit: Weber (Wb) / (T m<sup>2</sup>)



$\Phi$  = magnetic flux, Wb

B = flux density, T

A = component area normal to the field, m<sup>2</sup>

$\theta$  = angle between a normal to the surface and the field

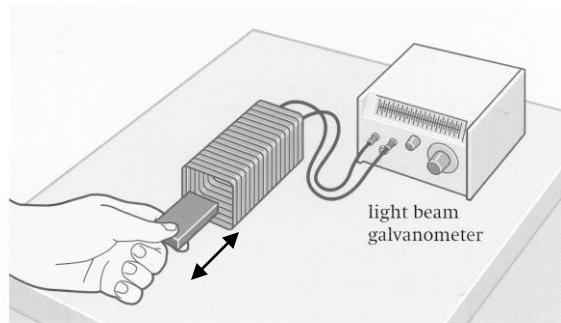
### Weber

\*\*Distinguish between magnetic flux density and magnetic flux.

### 2. Magnetic flux linkage

Magnetic flux linkage is defined as the product of the **flux  $\Phi$**  and the **number of turns  $N$** .

### 3. Laws of Electromagnetic Induction



A moving magnet near a coil generates a small current.

#### **Demo : Coil, Galvanometer, magnets resistor and connecting cables.**

- State the direction of current when the magnet enters the coil.
- Is there a current flow when the magnet is stationary in the coil?
- What can be observed when the magnet bar is pulled out from the coil?
- What are the changes if you were using two magnet bars?
- How does the observation changes if the coil is moved towards and away from the magnet?
- Why do we need to fix a resistor in series with the galvanometer?

Outcome:

1. Why there is emf induced?
2. Why the direction of induce emf changes?
3. What are the factors effecting the magnitude of induce emf?

Factors affecting magnitude of induced e.m.f. :

1. Magnetic flux density (B)
2. Cross sectional area of coil (A)
3. Number of turns of coil (N)

<http://www.youtube.com/watch?v=bkSsgTQOXVI&NR=1>

### Faraday's law of electromagnetic induction:

Magnitude of induce e.m.f. in conductor is directly proportional to the rate of change of magnetic flux.

$$E = N \frac{\delta \Phi}{\delta t}$$

### Lenz's law:

The direction of induced current / e.m.f. in a conductor is such that it opposes the cause of it.

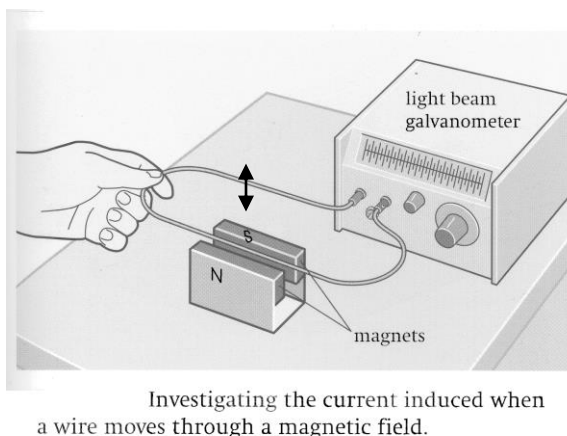
Combination of Faraday's and Lenz's law,

$$E = -N \frac{\delta \Phi}{\delta t}$$

$$\begin{aligned}\Phi &= BA \cos \theta \\ &= BA \cos \omega t\end{aligned}\quad \theta = \omega t$$

Therefore:

$$\begin{aligned}E &= -N \frac{\delta (BA \cos \omega t)}{\delta t} \\ &= -BAN \frac{\delta (\cos \omega t)}{\delta t} = -BAN\omega \sin \omega t\end{aligned}$$

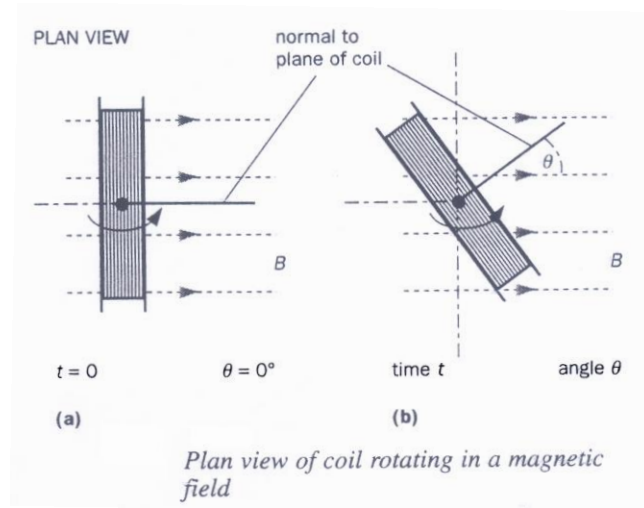
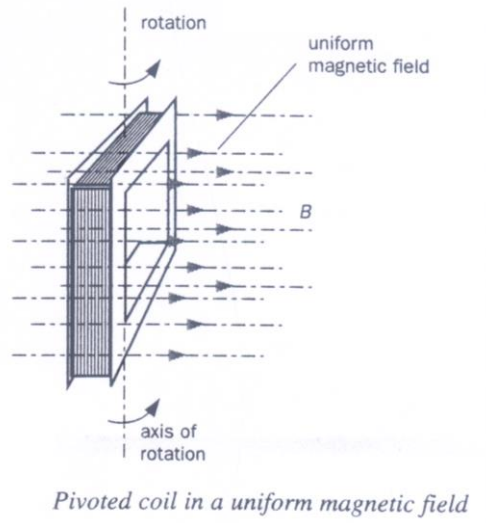


- How to determine the direction of the induced current in the conductor?

How energy is conserved in the above situation?

Which law is used to explain this?

#### 4. E.m.f induced in a rotating coil



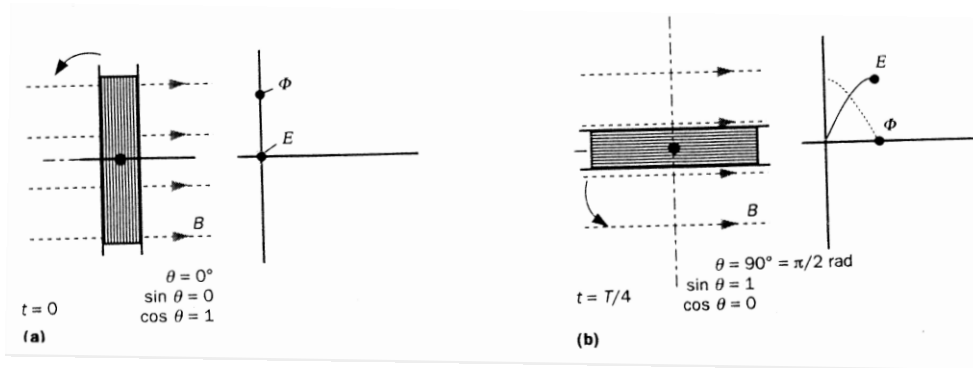
The flux link in each turn is given by:

$$\Phi = BA \cos \omega t$$

To calculate the induced e.m.f,  $\mathbf{E}$  :

$$\mathbf{E} = -N \frac{\delta (\mathbf{BA} \cos \omega t)}{\delta t} \quad (\text{Faraday's Law and Lenz's Law})$$

$$= -BAN \frac{\delta (\cos \omega t)}{\delta t} = BAN\omega \sin \omega t$$

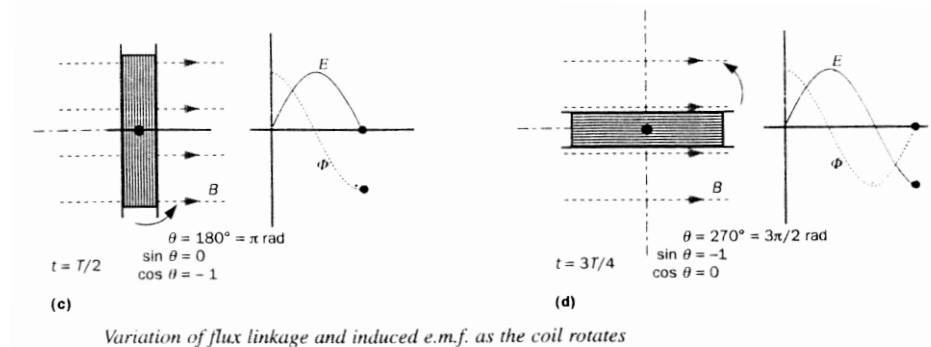


$$\Phi = BAN \cos \omega t = BAN$$

$$E = BAN\omega \sin \omega t = 0$$

$$\Phi = BAN \cos 90^\circ = 0$$

$$E = BAN\omega \sin \omega t = BAN\omega$$



$$\Phi = BAN \cos \omega t = -BAN$$

$$E = BAN\omega \sin \omega t = 0$$

$$\Phi = BAN \cos 270^\circ = 0$$

$$E = BAN\omega \sin \omega t = -BAN\omega$$

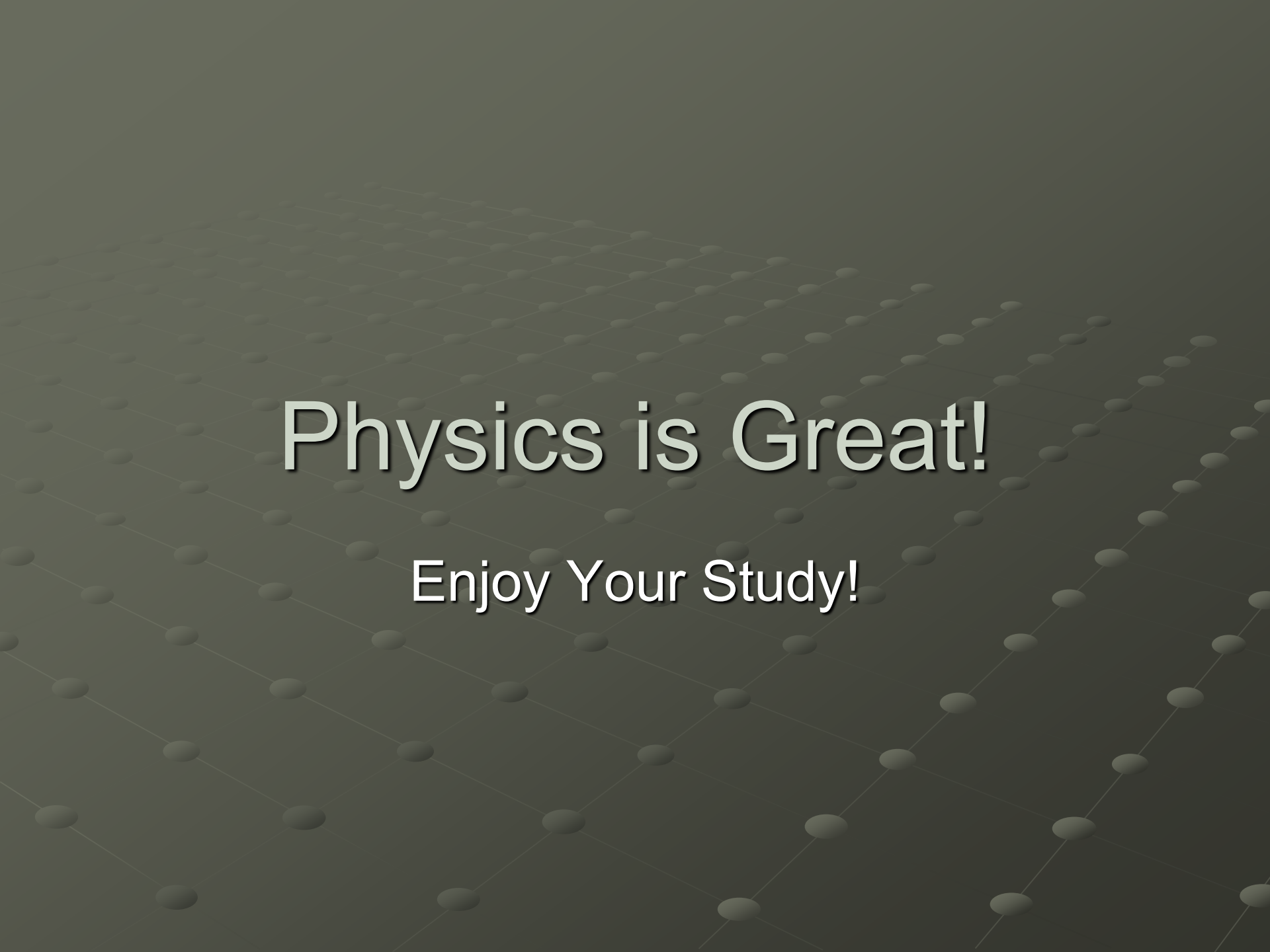
## 5. APPLICATIONS of Induction

### Back EMF in electric motors

You may have noticed that when something like a refrigerator or an air conditioner first turns on in your house, the lights dim momentarily. This is because of the large current required to get the motor inside these machines up to operating speed. When the motors are turning, much less current is necessary to keep them turning.

One way to analyze this is to realize that a spinning motor also acts like a generator. A motor has coils turning inside magnetic fields, and a coil turning inside a magnetic field induces an emf. This emf, known as the back emf, acts against the applied voltage that's causing the motor to spin in the first place, and reduces the current flowing through the coils. At operating speed, enough current flows to overcome any losses due to friction and to provide the necessary energy required for the motor to do work. This is generally much less current than is required to get the motor spinning in the first place.

If the applied voltage is  $V$ , then the initial current flowing through a motor with coils of resistance  $R$  is  $I = V / R$ . When the motor is spinning and generating a back emf, the current is reduced:  $I = (V - \epsilon) / R$   $\epsilon$ , the back emf, is usually a large fraction of  $V$ .



# Physics is Great!

Enjoy Your Study!